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AI ready: how to prepare archives for Artificial Intelligence, improve discoverability, and enhance access

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ABSTRACT
Artificial Intelligence (AI) and machine learning offer significant potential for archives: they can help identify sensitive content at scale, support the creation and enhancement of descriptive metadata, and enable new forms of access and interpretation. Yet AI also risks compounding long-standing challenges around archival description, bias, and power, especially where collections are fragmentary, under-described, or documented using outdated or harmful terminology.

This article asks: how can archives prepare their collections for AI in ways that enhance discoverability and access, while remaining grounded in archival principles and ethical commitments? Co-authored by a computer scientist and a digital humanist with experience in cultural heritage, the article offers a cross-disciplinary literature review and practice-oriented guidelines for "AI data preparedness" in archives. Drawing on work in data documentation, data readiness, collections-as-data, and archival theory, we argue that concepts such as provenance, appraisal, authenticity, reliability, and trustworthiness should shape AI applications in records and archives.

We distinguish between task-specific AI systems and generative AI, and propose workflows in which AI assists rather than replaces archivists, including description, terminology mapping, and sensitivity review. AI readiness, we argue, is an ethical and interventionist approach to data curation.

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Introduction

In the Libraries, Archives and Museums (LAM) sector, entire digital collections remain inaccessible or underused, not only because of technical barriers such as legacy formats and incomplete metadata, but also because of legal, ethical, and contextual constraints. Born-digital and digitized collections that contain personal or sensitive information cannot simply be put online without prior review. Traditionally, such review relied on archivists' close reading and contextual knowledge – a labour-intensive practice that struggles to scale to contemporary volumes of digital records.

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At the same time, AI and machine learning have been proposed as powerful tools to address some of these pressures.[1] They can support large-scale sensitivity review, assist in creating and enhancing descriptive metadata, and provide new modes of access such as conversational interfaces over collections. However, as archival scholarship has long argued, archives are not neutral data stores. They are shaped by power, selection, and silence.[2] Description and finding aids, while essential for access, are themselves historically contingent and implicated in exclusion and misrepresentation.[3] Community and Indigenous archives have further shown how archival practices can either reproduce or resist structural injustices.[4] Any application of AI to archives must therefore reckon with this intellectual and ethical legacy.

In this article, we argue that, under these conditions, automation is not a simple solution to archival challenges but a constrained necessity. Certain tasks – such as the initial pass over very large born-digital series, or the generation of draft item-level summaries – can no longer be carried out exclusively through manual methods. AI can assist archivists in undertaking this work, but only when collections are prepared and contextualized in ways that respect archival principles such as provenance, original order, and respect des fonds, and when its use is informed by critical perspectives on data, bias, and power.[5]

Our starting point is therefore not 'AI searching for a problem,' but a set of longstanding archival challenges that AI might help address if collections are made 'AI-ready.' By AI data preparedness, we mean the systematic assessment, consolidation, and documentation of archival data for use in AI systems, in ways that surface rather than obscure the scope, gaps, and limitations of collections. While some elements of this preparedness overlap with long-standing work on digital access and interoperability (e.g. format normalization, item-level metadata, and robust contextual description), we show that AI introduces distinct risks and requirements – particularly around decontextualisation, bias, privacy, and accountability – that call for explicitly interventionist and ethically grounded approaches to data curation.

This article contributes a practice-oriented framework for archival AI data preparedness that links archival principles to contemporary approaches in data documentation and readiness, and translates them into actionable guidance. Section 1 reviews scholarship at the intersection of archival studies, AI, and data curation (including InterPARES Trust AI). Section 2 distils this into four preparedness priorities – documenting completeness and excluded data; strengthening metadata and access (including narrative context); establishing coherent data types, formats, and file structures while preserving provenance; and defining evaluation metrics. It also illustrates their implications through workflows for AI-assisted description, sensitivity review, and access under human oversight.

Section 1: literature review

The interest in AI applied to the cultural heritage sector has increased significantly in the past five years. Crossing multiple disciplines and sectors, this scholarship has aimed to find solutions to the masses of digital data that are difficult to discover, access and analyse.[6]

Datasheets and documentation for archival data

Datasheets have emerged as a foundational framework for documenting datasets, addressing transparency, accountability, and ethical concerns in AI. In a pioneering 2021 article, Gebru et al. expanded the long-used practice of datasheets for datasets by proposing that such documentation should not only record technical attributes but also collection methods, recommended uses, and potential biases. Their aim was to 'increase transparency and accountability within the machine learning community, mitigate unwanted societal biases in machine learning models, facilitate greater reproducibility of machine learning results, and help researchers and practitioners to select more appropriate datasets for their chosen tasks.'[7] This work builds on longer research data management traditions in scholarly communication and reproducibility,[8] and has inspired adaptations for digital cultural heritage.

This framework has been adapted for digital cultural heritage by Alkemade et al., who emphasize the role of detailed metadata and provenance records in safeguarding the integrity and usability of archival datasets. For the authors, datasheets 'offer the advantage of allowing information to be collected in both a structured manner, whenever possible, and in a narrative form, whenever necessary.'[9] In deploying datasheets for cultural heritage data, the authors were influenced by the Collections as Data movement, which encouraged 'computational use of digitised and born digital' cultural heritage collections by making these collections available as data, 'amenable to computation.'[10]

Luthra and Eskevich expand on this concept with the idea of 'data-envelopes,' proposing a multidimensional approach to dataset documentation.[11] Their model integrates technical, contextual, and ethical dimensions, enabling institutions to address diverse stakeholder needs while enhancing dataset usability.

Data Readiness and assessment tools

Documenting datasets is a first step, complemented by data readiness approaches. *Data readiness* is a pivotal concept for preparing archival collections for AI, encompassing the technical, contextual, and quality aspects of data preparation. AIDRIN (AI Data Readiness Inspector), developed by Hiniduma et al. in 2024, offers a quantitative tool for assessing data readiness.[12] By evaluating parameters such as completeness, consistency, and usability, AIDRIN provides actionable insights for institutions aiming to enhance their archival collections' AI compatibility. In a 2025 article, Hiniduma et al. provide a comprehensive survey of data readiness frameworks, highlighting the challenges faced by organizations in aligning archival data with AI requirements.[13]

Data curation and ethical considerations

Effective data curation lies at the heart of preparing archival collections for AI. Bhardwaj et al. propose an evaluation framework for machine learning data practices through a data curation lens, emphasizing the importance of data quality, contextualization, and stakeholder engagement. Their work underscores the interplay between technical rigour and ethical responsibility in dataset creation.

Ethical considerations are also at the core of Jo and Gebru's paper, which offers strategies for collecting sociocultural data in machine learning, drawing lessons from archival practices.[14] They advocate for inclusive and context-aware approaches that prioritize the perspectives of marginalized communities. Similarly, Denton et al. provide a critical history of ImageNet, highlighting the ethical implications of dataset creation and use in AI.[15]

Critical perspectives on datafication and infrastructure

The datafication of cultural heritage collections presents both opportunities and challenges. Belteki et al. critique the underlying assumptions and power dynamics in data documentation, cataloguing, and sharing practices.[16] Their analysis calls for a reimagining of data infrastructures to support equitable and sustainable data use.

This focus on datafication and infrastructure is also central to Dodge et al.'s article, which offers insights from their documentation of large web text corpora. It provides lessons on scale, bias, and quality control.[17] These findings are particularly relevant for archival collections, where data heterogeneity and provenance pose significant challenges.

Archives, power, and the implications of 'collections as data'

As archival theorists have shown, these datafication processes are never neutral. Indeed, archives are the outcome of continuous processes of creation, selection, arrangement, and description. Early twentieth-century debates, often framed through the work of Jenkinson and Schellenberg, articulated enduring tensions between passive custodial models and more interventionist approaches to appraisal and arrangement.[18] Later scholarship has extended these debates, showing how archival practices participate in the construction of social memory and the distribution of power.[19]

One key theme is archival silence: the ways in which records and their descriptions reflect and reproduce the perspectives of dominant actors while marginalizing or erasing others.[20] Descriptive tools such as finding aids have traditionally been treated as gateways to collections, but they also embody archival choices and assumptions; Wiedeman warns that these tools can obscure gaps, flatten complexity, or mislead users if treated as transparent representations of archival holdings.[21] Community archives and activist projects have responded by foregrounding community participation, affect, and accountability in archival work.[22] Indigenous scholars and archivists have similarly argued for community-based provenance and control over records.[23]

These perspectives are directly relevant to AI applications in archives. AI systems trained on existing archival descriptions and catalogues risk inheriting and amplifying their biases and silences. For instance, systems that learn from racist or colonial subject headings may reproduce harmful language in generated descriptions or retrieval outputs, thereby re-inscribing historical violence.[24] Similarly, models that rely on incomplete or uneven documentation can misrepresent the scope or significance of collections, presenting partial holdings as comprehensive.

The 'collections as data' movement seeks to make digital cultural heritage materials available for computational use, encouraging LAM institutions to provide their holdings in machine-actionable formats.[25] For archives, this shift can be valuable, enabling new forms of analysis and engagement. However, as Caswell reminds us, 'the archive' in abstract computational discourse is not the same as 'archives' as institutions, records, and practices.[26] Treating archives exclusively as data risks flattening their complex provenance, legal status, and social meanings. In this article we therefore adopt a more cautious framing of *collections as/for data*: we acknowledge that digital surrogates and descriptive metadata can be structured for computational purposes, while insisting that such structuring must remain accountable to archival principles and to the communities whose lives and histories are represented in the records.

AI applications

Archival collections have immense potential for AI/machine learning applications, from training models to generating insights for cultural heritage studies. Desai et al. explore the role of pretraining data in archival contexts, highlighting the importance of curating datasets that are representative, diverse, and ethically sourced.[27]

This ethical focus is shared by Hutchinson et al., who discuss accountability practices for machine learning datasets, drawing parallels to software engineering and infrastructure management.[28] Their recommendations include implementing robust documentation practices and fostering cross-disciplinary collaboration to address biases and gaps in archival collections.

Archival principles for AI applications are central to the InterPARES project which has, over several phases since 1999, developed theoretical and methodological frameworks for the long-term preservation of authentic electronic records.[29] Its most recent phase, InterPARES Trust AI ('I Trust AI'), focuses explicitly on Artificial Intelligence and records/archives. The project aims to identify AI technologies that can address critical archival challenges, determine the benefits and risks of applying AI to records, ensure that archival concepts and principles inform AI development, and validate these approaches through case studies and demonstrations.

InterPARES Trust AI has highlighted, among other points, that AI should be adopted in archives not as a generic technology in search of applications, but as a set of tools explicitly designed around archival requirements such as authenticity, reliability, chain of preservation, and accountability. Preliminary results show both the promise and the constraints of AI in areas such as declassification review, classification and appraisal support, and automated assistance for description. In all cases, the project stresses the need for archivists' conceptual frameworks to drive system design, and for transparency, documentation, and auditability to remain central.

Our approach in this article is aligned with these findings. By focusing on AI data preparedness, we aim to complement InterPARES Trust AI's work on principles, governance, and system design. We focus primarily on the collection-level and dataset-level preconditions – completeness, metadata, structure, documentation, and ethical awareness – without which AI tools cannot operate responsibly or effectively in archival contexts.

Future directions

The preparation of archival collections for AI is an evolving field, with emerging trends pointing towards greater automation, interoperability, and ethical accountability. Fernandes et al. review technological advancements in data preparation, emphasizing the need for scalable and adaptive tools that can accommodate the unique characteristics of archival data.[30]

To integrate AI into archival practices effectively, institutions must adopt a holistic approach that balances technological innovation with cultural and ethical considerations. This includes investing in training, fostering collaboration between archivists and technologists, and developing policies that prioritize inclusivity and sustainability.

In other words, preparing archival collections for AI applications requires a multifaceted approach that addresses technical, contextual, and ethical dimensions. By leveraging frameworks like datasheets and data-envelopes, adopting tools for data readiness assessment, and embracing inclusive curation practices, institutions can enhance the value and impact of their collections in the AI era. Our article aims to continue to explore the intersections of technology, culture, and ethics, ensuring that archival collections remain accessible, representative, and accountable in a rapidly changing digital landscape.

Section 2: AI readiness guidelines

For professionals in Libraries, Archives and Museums, the prospect of applying AI to archival collections often raises two intertwined questions. First, which archival problems can AI realistically help address, given the complexity, sensitivity, and incompleteness of many collections? Second, what conditions must be in place for AI applications to be trustworthy, accountable, and aligned with archival principles?

In this section we answer these questions by translating the data readiness frameworks discussed above into practical guidelines for archival contexts. We distinguish between:

- Task-specific AI, where models are trained to perform a well-defined function such as classifying record types, detecting named entities, or ranking search results; and
- Generative AI, particularly Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) systems, which can produce natural-language summaries, explanations, or narratives grounded in archival collections.

While both families of systems require careful preparation of training and reference data, their requirements differ. Task-specific systems tend to benefit most from structured, labelled, and representative datasets. Generative systems are more flexible with respect to input structure, but depend heavily on rich contextual and narrative metadata, robust provenance, and comprehensive documentation of collection scope and gaps.

One particularly effective generative AI approach relevant to Libraries, Archives and Museums is Retrieval-Augmented Generation (RAG), which couples retrieval from a curated collection with text generation. By grounding outputs in retrieved items, RAG reduces the risk of invented sources and supports traceable links back to records and metadata. In archival terms, it can help keep outputs faithful to authentic records (in the sense of integrity), without implying that those records are complete, objective, or culturally unproblematic; interpretation, critique, and contextualization remain the responsibility of archivists and users.

In what follows we describe how AI data preparedness can be achieved in four key areas – completeness, metadata and access, data types and formats, and application-specific metrics – and show how these areas underpin concrete AI-assisted archival workflows.

Framework

Established frameworks focused on data preparedness already exist. For example, Hutchinson et al. cast a parallel between the software development lifecycle and a proposed dataset development lifecycle.[31] The lifecycle model considers development as a non-linear but cyclical engineering project. It establishes a set of phases: dataset requirements specification, design, implementation, testing, and maintenance. The authors emphasize the importance of establishing 1) dataset documentation practices throughout the lifecycle of a dataset, 2) oversight processes, including reviews and audits, and 3) robust maintenance mechanisms, including updates and improvements of the dataset. All these points are relevant to LAM organizations, as we will further discuss below.

In Hiniduma et al. (2025), the authors separate their discussion of data preparedness along the lines of structured data and unstructured data, this latter category is further divided in text and multimedia.[32] Structured data is organised in a consistent format, it follows a specific schema and is usually organised in spreadsheets and relational databases (i.e. tables), or other suitable key-value formats (e.g. JSON). Most publications about data preparedness for AI discuss the case of structured data, which is also the least common in LAM collections. When discussing unstructured data, most of the attention has been devoted to metrics and methods to assess data quality. Nevertheless, while important, data quality is but one among several aspects to consider when discussing data preparedness for AI applications in LAM collections. In what follows, we start by detailing and expanding the framework of AI data readiness categories proposed in Hiniduma et al.[33] and consolidated in Hiniduma et al. [34]

Quality

- **Completeness**: Assesses the presence of all available data with respect to the original collection, or a representative sample thereof.
- **Outliers**: Identifies anomalous data points or items. For example, empty documents after text recognition.
- **Duplication**: Detects and removes possible duplicates (with human oversight).
- **Data preparation practices**: Assesses the robustness of the methods used for preparing data. For example, the digitization pipeline.
- **Timeliness**: Evaluates whether data is current and relevant.

Data quality is important for AI applications. Ensuring that born-digital or digital surrogates of cultural items and collections used to train or inform task-specific and generative models are accurate, representative, and authentic is critical. High-quality comprehensive digitization, minimal duplication, and carefully managed outliers ensure generative models reflect the cultural artefacts as represented in the collection, rather than data distortions.

Understandability and usability

- **Metadata availability and quality**: Ensures that comprehensive metadata is present at collection and, if possible, item level. Metadata uniformity and depth, to the extent possible, are more important than the specific schema that is used.
- **Provenance**: Tracks and documents the origin and lineage of data.
- **Interfaces for data access**: Ease of data access and interaction, including following the FAIR principles.[35] Programmatic data access is highly desirable, for example via Application Programming Interfaces (APIs).

For generative AI applications in LAM contexts – especially when using RAG – metadata and well-documented provenance are central to retrieval, access control, and contextual interpretation. Item-level metadata improves targeting and performance at scale, and narrative context (e.g. scope and content notes) can enrich outputs; we develop these points further in the Guidelines section below. Ideally, both metadata and content are accessible via programmatic interfaces such as APIs.

In cultural heritage collections, data often come in multiple languages. At the National Library of Wales, for example, AI models would need to work well on Welsh-language archival data. While AI has long focused mainly on the English language, new models have been trained on multiple languages – opening new opportunities for LAM professionals to enhance the discoverability and access to their multilingual collections.[36]

Structure and organisation

- **Used data types**: Appropriateness and consistency of data types.
- **Quality of data schema**: Design and structure of data schema.
- **File format and storage system:** Uniformity and suitability of file naming, file structure, formats, and storage systems.
- **Data access performance**: Efficiency in retrieving data.

Generative AI models are very apt at using unstructured inputs as well, especially texts without a clearly defined schema, such as complex documents stored in human-readable formats such as doc or pdf. However, structured data schemas and interoperable formats facilitate training task-specific models and using data for generative AI applications. Uniformity in file naming, folder structure, and formats remains critical to allow models to select relevant documents. Generative AI applications benefit from structural consistency, all the while remaining considerably more flexible with respect to the need for adopting uniform data schema.

This ideal of uniformity can seem hard to reconcile with the real-world complexity of born-digital collections. As we have seen in the introduction, born-digital collections inherently arrive in diverse, uncontrolled, and often non-standard formats. This diversity is an essential part of their archival context. Uniformity desired for AI preparedness typically results from digital preservation workflows and interventions rather than original archival acquisition.

While this article emphasizes uniformity of formats and removal of duplicates for AI readiness, it is also important to highlight that a key archival principle, particularly for born-digital materials, is maintaining provenance. Original file names, formats, dates of creation, and amendments constitute critical provenance information for archives. These are core tenets of *respect des fonds* – an archival theory developed in 19th-century Europe and still influential today. Any automated processes (e.g. removing duplicates or reformatting) should never compromise this provenance. In other words, AI applications should not automatically enforce alterations without some human control. What we suggest is not to alter originals directly, as this would be incompatible with *respect des fonds*. Instead, we recommend treating the acquired files as preservation copies and creating separate access or dissemination versions – akin to Dissemination Information Packages (DIPs) in the Open Archival Information System (OAIS) model – with an emphasis on uniform data types and formats for efficient AI deployment.

Value

- **Feature importance**: Significance of dataset features.
- **Labels**: Availability and accuracy of labels for supervised learning.
- **Data point impact**: Influence of individual data points.
- **Uncertainty in data**: Measurement of data confidence.

In LAM AI applications, value as defined above is less applicable because fewer datasets are structured and numerical, whilst instead most are unstructured and textual or multimedia. Therefore, collection data is seldom structured enough for features, labels, data points, and uncertainty to be present or clearly measured. This is even less of a concern for generative AI applications, due to their capability to process unstructured data natively.

Fairness and bias

- **Class imbalance**: Balance of data categories.
- **Class separability**: Distinctness between classes.
- **Discrimination index**: Identification of potential biases.
- **Population representation**: Diversity and representativeness.

LAM collections often contain gaps and selections or reflect historical perspectives. For example, colonial historical records are often very sensitive, especially when they show violence and humiliation of colonized populations. Even when collections have been digitized, they are not always easily discoverable, for instance in the case of missing or problematic metadata containing racist or outdated language. Making archival records more accessible, in a responsible way, is a key priority.

Class imbalance and separability are only a concern for settings where data classes are salient. The identification of potential biases should also be defined situationally in view of the specificities of the collection and the intended use cases. A clear definition of what constitutes bias and how it might impact an AI application should be stated before assuming bias is present, and actions are taken. Lastly, population representation is to be considered whenever a collection is not fully available, but just a sample of it, or when excluded data is to be considered. Beyond overt slurs or explicit hate speech, many of the most pervasive biases in archival records operate at the level of worldview: assumptions about what counts as 'civilised,' 'proper,' or 'normal,' and about gender, sexuality, race, and class. These normative frameworks are woven into language and description in ways that are far harder for both humans and AI to detect and contest. There is currently no consensus within the archival sector on how best to address such deep-seated biases; documentation, critical reflection, and community engagement therefore remain essential complements to any technical mitigation strategy.

Governance

- **Collection**: Ethics, sampling methods, regulatory compliance.
- **Processing and curation**: Data anonymization and curation.
- **Application**: Restrictions on data usage and bias evaluation in data analysis.
- **Security**: Access controls and sharing protocols.
- **Privacy**: Evaluation of data privacy.

AI applications in LAM must navigate sensitive ethical, legal, and privacy considerations carefully. Good governance assures ethical generative applications, data protection, and responsible usage of cultural heritage. These aspects of data preparedness are beyond the scope of this article.

AI application-specific metrics

Model-specific metrics

Evaluates metrics specific to the AI models and their intended applications, ensuring the data meets the requirements for successful model training and deployment. Users can define data metrics that are specific to the AI model they are developing. In generative AI applications for LAM collections, examples include the evaluation of the accuracy of the retrieved information and its completeness. (See Table 1 for a summary of AI data preparedness categories for LAM collections).

Guidelines

The list of data preparedness categories proposed by Hiniduma et al. (2024, 2025) is relevant to varying degrees to LAM organisations seeking to develop AI applications.[37] Based on our experience with LAM collections, we consider it important to underline a few categories that are particularly significant, either because of the specificities of LAM collections and organisational aims, or because they entail a shift in approach when considering generative AI rather than task-specific AI applications. These are: 1) Data completeness (under Quality); 2) Metadata and access (Understandability and usability); 3) Data types, formats, and file system structure (Structure and organization); and 4) AI application-specific metrics. We discuss them in turn providing more context and general guidelines on how to approach them.

Table 1. Summary of AI data preparedness categories for LAM collections

Category	Importance for LAM applications
Quality	Essential for reliable outputs in any AI application
Understandability and usability	Critical for any application, and in particular for contextually accurate generative modelling.
Structural quality	Facilitates training, generative consistency and completeness. While generative AI can use unstructured data, consistent file naming, structure, formats remain important.
Value	Ensures cultural relevance and appropriateness of generated outputs. More relevant for structured data, thus less applicable in a LAM setting.
Fairness and bias	Prevents distortions and omissions in generated outputs, or lack of awareness of learned model limitations.
Governance	Supports ethical use, privacy protection, and regulatory compliance. Of general importance no matter what the intended scenario of data usage is.
AI application-specific	Allows for the tailored assessment for generative and task-specific LAM use cases and collections.

Data completeness is a critical aspect of preparing cultural or LAM collections for generative AI applications. Completeness refers explicitly to the extent to which the dataset encompasses all available or historically significant items and features. It is crucial to document clearly – ideally in detailed, collection-level metadata – the degree of completeness of the collection. This documentation should clarify whether the dataset represents the entirety of the available material or whether it has been selectively sampled, curated, or only partially digitized from a more extensive original collection.

Explicitly stating completeness involves explaining the rationale behind the selection criteria, identifying gaps in the dataset, and contextualizing what content might be missing, unavailable, or deliberately excluded. For instance, if certain periods, creators, cultures, or media types within the collection have not been digitized or are only partially represented, these gaps and omissions must be explicitly noted. Such detailed metadata significantly enhances the transparency of the dataset and informs users or AI models about its scope, boundaries, and inherent biases.

In generative AI applications specifically, completeness plays a pivotal role in ensuring outputs are contextually accurate, culturally coherent, and historically faithful. Clear contextual information about the dataset's completeness enables generative models to account for and acknowledge limitations explicitly, producing outputs that reflect the dataset's nature authentically. To enrich generative models further, additional contextual materials such as interpretive essays, expert commentary, curatorial statements, or secondary literature about the collection can serve as valuable supplementary input or training data. Such resources help generative models to build a deeper understanding of the historical, cultural, and interpretive frameworks underlying the data. Consequently, the model can generate more meaningful, relevant, and contextually nuanced outputs.

In contrast, data representativeness – the extent to which data accurately captures the diversity and variability of the broader category or phenomenon it aims to represent – takes on greater significance for traditional task-specific AI applications. Representativeness is particularly crucial when models must generalize effectively, performing well not only on previously encountered examples but also on new, unseen, yet similar data points. Ensuring representativeness allows these task-oriented models to avoid overfitting and perform robustly across a wide array of contexts. Thus, while generative AI benefits most from explicitly clarifying and documenting data completeness to produce authentic, context-aware outputs, task-specific AI applications benefit primarily from prioritizing representativeness to achieve reliable generalization and adaptability.

Metadata – structured information describing data – is a foundational component of effective data management and is crucial across all AI applications, especially within cultural and LAM settings. High-quality, comprehensive metadata enhances the discoverability, accessibility, interpretability, and interoperability of datasets. It empowers users and AI systems alike to understand the nature, origin, and context of individual data items, thereby facilitating accurate, meaningful, and responsible analyses.

Archival descriptive standards such as Dublin Core, CIDOC CRM, MARC21, and ISAD(G) deliberately combine highly structured elements (names, dates, places, identifiers) with narrative fields such as biographical or administrative history, scope and content, and custodial history. These narrative components have long been recognized as essential for human users, providing multi-paragraph accounts of the context, content, and significance of records. Our argument is therefore not that archival metadata lacks narrative, but that these rich narrative elements – and additional curatorial commentary – can also be written and maintained with machine consumption in mind.

In the context of generative AI, item-level metadata becomes particularly valuable. Even when digital files are machine-readable, models often rely on metadata to:

- filter and restrict retrieval to relevant subsets of a collection (e.g. by date range, creator, record type, language);
- compensate for poor OCR, complex layouts, or non-textual media where full-text analysis is unreliable or impossible; and
- provide concise, structured cues that guide retrieval and reduce computational cost when working at scale.

Item-level metadata therefore plays a complementary role: AI systems may still 'read' the full document where appropriate, but they depend on metadata to target that reading effectively and to align their behaviour with archival arrangement and access conditions. Generative AI applications also invite us to make more systematic use of what we call discursive metadata: narrative, free-form textual documentation that includes both existing archival fields (e.g. scope and content notes, biographical histories, custodial histories) and additional interpretive or curatorial texts. These may include exhibition catalogues, research reports, community-authored statements, or critical essays that foreground multiple perspectives on the records. Such discursive metadata provides richer contextual information, detailed background stories, and nuanced discussions surrounding individual items or collections.

Generative models, particularly LLMs integrated within RAG frameworks, are well suited to parsing and leveraging extensive unstructured text. By providing discursive metadata alongside structured elements, institutions enable generative models to synthesize information in ways that more accurately reflect the complexity, ambiguity, and layered interpretations characteristic of archival collections. The advantage of these models is that they can work with both structured schemas and free-form text describing a collection or item in any level of detail deemed necessary, reducing the need to force all information into rigid, highly constrained fields when creating metadata. Narrative and discursive metadata, when thoughtfully curated and documented, can thus significantly enrich the outputs of generative AI while remaining grounded in archival principles.

The choice of data types, formats, and file system structure significantly influences the effectiveness, efficiency, and accuracy of AI applications, particularly within cultural and LAM contexts. AI systems require clarity and coherence in how data is organized, identified, and interconnected. A logical, consistent, and clearly documented file structure, alongside thoughtful and uniform naming conventions, is therefore paramount.

When preparing a dataset for AI applications – particularly in cases where collections have not been born digital but rather digitized from physical originals – establishing clear and traceable relationships between digital files and their physical counterparts is essential. The file system structure and file naming conventions should explicitly reflect these relationships, facilitating direct mapping between digital surrogates and original artefacts. Such practices ensure transparency and reproducibility, enabling future users and AI models to reliably track data provenance, verify authenticity, and appropriately interpret results.

Moreover, file system structures must also clearly convey internal relationships among files themselves. For example, related objects or multiple representations of the same object – such as different resolutions or formats – should be stored in systematically structured directories with coherent and consistent naming. These strategies enable AI models to easily detect and leverage relationships among items, supporting richer contextual understanding, enhanced data retrieval, and more accurate processing outcomes.

In addition to a robust file organization, uniformity in data types and formats within collections is important for achieving consistent and reliable AI results. When datasets comprise a diverse mixture of file formats (such as varying image formats like TIFF, JPEG, PNG, or different text encodings and document types like PDF, XML, TXT), AI models are forced to employ multiple, possibly inconsistent parsers or processing workflows, each potentially introducing discrepancies or varying levels of accuracy. These inconsistencies can introduce unintended biases, data loss, or errors into the analysis, ultimately degrading the quality, comparability, and reliability of AI outputs. Thus, wherever feasible, collections intended for AI applications should standardize data types and formats according to recognized, widely-supported, and AI-friendly standards. For textual data, this might involve standardizing documents in plain text, UTF-8 encoded XML, or JSON formats. For visual data, this might mean standardizing on widely used formats like JPEG or TIFF, depending on use-case requirements. Similarly, audiovisual data benefits from consistent codecs and container formats, simplifying downstream parsing and processing. This uniformity significantly enhances data interoperability, streamlines processing pipelines, reduces technical complexity, and improves the overall performance and interpretability of AI results.

Therefore, careful attention to establishing coherent file system structures, clear and logical file naming conventions, transparent mapping to physical originals, and uniformity in data types and formats is essential. Together, these strategies create a robust foundation that empowers AI applications – both traditional task-specific models and more advanced generative AI – to achieve accurate, reproducible, and contextually meaningful results, fully leveraging the potential embedded in rich digital cultural heritage collections.

AI application-specific metrics play a crucial role in assessing and validating AI systems within Libraries, Archives and Museums. Unlike generic or broadly applicable metrics, application-specific metrics offer targeted and meaningful measures of success, directly aligned with the distinct objectives, priorities, and audiences associated with each AI-driven project.

Establishing these tailored metrics begins by clearly defining the project's goals, expected outcomes, and intended target users. Such goals and users vary widely depending on the project context: for instance, an AI application designed to enhance visitor engagement in a museum context may focus on user satisfaction, interpretive accuracy, or visitor dwell time, while an archival retrieval tool may prioritize accuracy, comprehensiveness, or efficiency of information discovery. Clearly articulating these objectives early in the project allows LAM institutions to identify precisely which outcomes are most significant, ensuring that the evaluation approach is relevant, robust, and directly informative to stakeholders.

Accompanying these metrics, a structured and transparent evaluation protocol must also be established. Such a protocol systematically outlines how each metric will be measured, under which conditions, and with which data. It may incorporate quantitative methods (e.g. automated accuracy checks, precision-recall computations), qualitative assessments (e.g. expert validation, user interviews, usability testing), or a combination thereof. Having clearly documented procedures helps ensure that the evaluation process is consistent, repeatable, transparent, and reliable.

In the specific case of generative AI – particularly when it functions as an information retrieval or exploratory interface atop a cultural heritage collection – the choice of metrics must align closely with both the nature of the generative application and its anticipated use. For example, if generative AI is employed as a conversational interface for information retrieval from a digital archive, an evaluation framework might explicitly incorporate established information retrieval metrics such as precision (the proportion of retrieved items that are relevant) and recall (the proportion of all relevant items successfully retrieved). Evaluating these metrics ensures that the generative system not only produces fluent or engaging outputs but also reliably and consistently provides users with accurate, relevant, and comprehensive information from the dataset.

Further, given the inherent complexity and interpretive richness often encountered in LAM contexts, these standard quantitative metrics can beneficially be complemented by qualitative evaluation approaches. User satisfaction surveys, structured interviews, or expert assessments can reveal critical insights into interpretive accuracy, cultural sensitivity, usability, or perceived value of the generative outputs. By integrating qualitative measures alongside quantitative ones, LAM organizations can gain deeper, contextually informed insights, helping them assess the extent to which their generative AI solutions meet real-world user needs and organizational objectives.

Ultimately, the thoughtful definition and careful application of AI application-specific metrics – grounded explicitly in the goals, context, and audience of the AI initiative – provide LAM institutions with a framework for evaluation. This tailored approach ensures that AI systems align closely with organizational priorities, resonate with target users, and produce outcomes that are not merely technically successful, but culturally, socially, and institutionally valuable.

Best practices and viable solutions

Preparing data effectively for AI, especially generative AI, requires careful planning in Libraries, Archives and Museums. A first priority is to establish, as early as possible, a coherent approach to data types, formats, file naming conventions, and file structure. When digitization or ingest projects adopt consistent practices from the outset, later AI work becomes far easier: data pipelines are simpler, models behave more predictably, and outputs are easier to interpret and reproduce. Clear file naming and directory structures that explicitly link digital files to their physical or original counterparts support traceability and authenticity and can often be implemented programmatically as part of data preparation.

In parallel, institutions can begin to experiment with AI-assisted creation of item-level metadata using LLMs. Summaries, topics, and key entities (such as names and dates) can be extracted and stored in dedicated metadata fields that are much faster for an LLM to inspect at retrieval time than entire documents. This can substantially reduce the labour required for first-pass description, provided that outputs are subject to human review and that culturally sensitive or contested terminology is checked carefully.

Beyond structured metadata, LAM institutions should selectively integrate contextual and narrative material – curatorial notes, interpretive essays, exhibition catalogues, community statements, and relevant secondary literature – into the datasets prepared for generative AI. Embedding this discursive metadata directly into the AI-addressable corpus enables more context-rich and meaningful generated content, and provides an avenue for multiple perspectives to be represented alongside institutional description.

To make effective use of these possibilities, staff need opportunities to develop confidence and critical understanding of generative AI. Short training activities, workshops, and hands-on pilots can familiarize archivists and other LAM professionals with the capabilities and limitations of LLMs. Small-scale, time-bound projects are especially valuable: they allow institutions to test specific use cases, evaluate results, and refine workflows before committing to larger-scale deployments. Such pilots can also surface institutional, ethical, and technical issues that might otherwise remain invisible in purely speculative discussions.

AI-assisted archival workflows: description, sensitivity review, and access

The guidelines presented above might suggest that extensive metadata and highly standardized formats are strict prerequisites for any AI use. In practice, however, AI can play a meaningful role even when description is incomplete or uneven, provided its limitations are acknowledged and archivists remain in control of appraisal, interpretation, and final decision-making. Here we sketch three broad classes of AI-assisted workflows that are already emerging in archival practice.

Draft descriptive metadata and subject enhancement

Generative models can be used to propose draft descriptions for individual items or groups of records based on their content and existing contextual information. For example:

- extracting candidate titles, dates, and brief summaries from born-digital documents or digitized text;
- suggesting topical keywords based on controlled vocabularies, while flagging where no appropriate term exists;
- grouping items into candidate series or sub-series based on shared features (e.g. sender/recipient, time period, place).

Crucially, these outputs are proposals, not final descriptions. They can reduce the time archivists spend on first-pass description and help surface patterns in large corpora, but they require systematic human review, especially to ensure that proposed terms are accurate, culturally appropriate, and sensitive to community perspectives. Archival scholarship on the hazards of finding aids[38] and on archival silences[39] reminds us that description is always interpretive; AI-generated descriptions are no exception and must be treated as part of an ongoing descriptive dialogue, not as authoritative ground truth.

Sensitivity review and harmful content detection

Another promising but constrained application concerns the support of sensitivity review. AI systems trained on examples of personal data, legally restricted information, or harmful and hateful expressions can help:

- flag records likely to contain personal identifiers, confidential information, or protected categories;
- identify overt occurrences of racist, sexist, homophobic, or otherwise harmful language;
- surface segments that may require closer human scrutiny before online release.

These systems do not replace professional judgement nor resolve complex ethical and legal questions. However, they can help archivists prioritize attention within very large corpora and document the rationale for inclusion, redaction, or restricted access. Critical work on racist materials and 'archiving hate'[40] underscores the importance of not simply removing or hiding harmful records, but of contextualizing them; AI tools must be designed to support this nuanced practice, not to erase uncomfortable histories.

Access, discovery, and interpretive interfaces

Finally, RAG-based systems can act as conversational or exploratory interfaces over collections. When grounded in well-documented, AI-ready datasets, they can:

- provide users with natural-language summaries of complex records or series;
- offer bilingual or multilingual access to collections[41];
- surface relevant materials through semantic similarity rather than exact keyword match, potentially widening access for users unfamiliar with archival jargon.

Such systems must be carefully evaluated using application-specific metrics (e.g. precision, recall, user satisfaction) and monitored for hallucinations, over-confidence, or biased retrieval. They should clearly disclose their scope (which collections they cover) and limitations (what is missing or under-represented), reflecting the attention to completeness and excluded data discussed above. In line with InterPARES Trust AI, they should be designed to uphold archival notions of authenticity and trust, for example by linking generated answers explicitly back to the underlying records and descriptive metadata.

Across these workflows, AI remains a supporting actor. Its usefulness depends on the quality and transparency of the underlying data and on archivists' ability to frame tasks, interpret outputs, and decide how (or whether) to integrate them into institutional practices. AI data preparedness, as we understand it, is therefore not about preparing archives for AI as an external, neutral technology, but about preparing AI to work within the constraints, responsibilities, and possibilities of archival practice.

Approaching the end of this section, we touch upon other topics closely related to data preparedness. We discuss the importance of explaining data exclusion, the cultures of data curation for AI applications, dataset documentation best practices, and model auditing.

With a focus on the documentation of web-crawled corpora, Dodge et al. suggest the inclusion of metadata: *included* data and *excluded* data.[42] In the context of LAM collections, excluded data is particularly significant. Data might be missing (e.g. gaps in the collection), or might be excluded from usage for a variety of reasons (e.g. privacy). Some LAM collections aim for exhaustiveness (archives), others aim for representativeness (museums), therefore the cause and significance of excluded data is not only relevant, but also situational. Excluded data should be explained in as much detail as possible in datasheets or similar documentation artefacts. If possible, the cause of the omission should be clearly articulated. For example, an archival series might be missing some years due to historical reasons, curatorial selection, or legal constraints. The documentation should be human-readable in the context of task-specific applications, but the same documentation can also be used by LLMs in the context of generative AI applications. In the former case, the primary concern is to understand the generalization limitations of the AI applications being developed, as impacted by excluded data. In the latter case, excluded data will impact the answers of the system: users should be informed of the resulting limitations.

Conclusion

AI data preparedness in archives sits at the intersection of several ongoing developments: the maturation of digital preservation infrastructures, the 'collections as data' movement, renewed attention to archival power and silences, and the rapid diffusion of generative AI. This article's contribution is to show that preparing archival collections for AI is not simply technical optimization, but an inherently archival and ethical project – one that must preserve provenance, surface gaps and silences, and remain accountable to communities and users.

Bringing together archival scholarship with recent work on dataset documentation and data readiness (and in dialogue with InterPARES Trust AI), we propose a practical framework for making collections responsibly AI-ready. We highlight four priorities: (1) documenting completeness and excluded data; (2) strengthening metadata and access, including narrative context; (3) establishing coherent data types, formats, and file structures while preserving provenance; and (4) defining application-specific evaluation metrics and protocols.

We also outline concrete workflows in which AI can assist – rather than replace – archivists: generating draft descriptions and subject suggestions, flagging potentially sensitive or harmful content for review, and enabling new discovery and access interfaces grounded in records. These workflows are not universal solutions; their value depends on careful design, ongoing oversight, transparent documentation, and collaboration between archivists, technologists, and the communities represented in the records.

Future work should deepen and extend this collaboration through longitudinal studies of AI-assisted description, user studies of AI-mediated access, and more systematic exploration of model auditing in archival settings. It also requires institutional investment in skills and infrastructures so that archives can engage with AI on their own terms. By treating AI data preparedness as both a technical and a profoundly archival endeavour, we hope to contribute to a more grounded, responsible, and generative conversation about AI in archives.

Notes

1. Arias Hernandez et al., "Artificial Intelligence and Machine Learning."
2. Schwartz and Cook, "Archives, Records, and Power"; Carter, "Of Things Said and Unsaid"; and Caswell, "'The Archive' Is Not an Archives."
3. Wiedeman, "The Historical Hazards of Finding Aids."
4. Cifor et al., "'What We Do Crosses over to Activism'"; McCracken and Hogan, "Community First."
5. Noble, *Algorithms of Oppression*.
6. See, for examples: Colavizza et al., "Archives and AI"; Canning and Jaillant, "AI to Review Government Records"; Jaillant and Caputo, "Unlocking Digital Archives"; Jaillant, "Archives, Access and Artificial Intelligence"; Hernandez et al., "Artificial Intelligence and Machine Learning Competencies"; and Shinde et al., "AI in Archival Science."
7. Gebru et al., "Datasheets for Datasets."
8. See, for example, Briney et al., "Foundational Practices of Research Data Management."
9. Alkemade et al., "Datasheets for Digital Cultural Heritage Datasets." See also Pushkarna et al., "Data Cards."
10. Padilla et al., "Always Already Computational."
11. Luthra and Eskevich, "Data-Envelopes for Cultural Heritage."
12. Hiniduma et al., "AI Data Readiness Inspector."
13. Hiniduma et al., "Data Readiness for AI."
14. Jo and Gebru, "Lessons from Archives."
15. Denton, "On the Genealogy of Machine Learning Datasets."
16. Belteki, "Datafication and Cultural Heritage Collections Data Infrastructures."
17. Dodge, "Documenting Large Webtext Corpora."
18. Stapleton, "Jenkinson and Schellenberg."
19. Schwartz and Cook, "Archives, Records, and Power."
20. Carter, "Of Things Said and Unsaid"; and Caswell, "'The Archive' Is Not an Archives."
21. Wiedeman, "The Historical Hazards of Finding Aids."
22. Cifor et al., "'What We Do Crosses over to Activism'"; Nelson, "Archiving Hate"; and Eichhorn, "Beyond Digitisation."
23. McCracken and Hogan, "Community First."
24. Noble, *Algorithms of Oppression*.

25. Padilla et al., "Always Already Computational."
26. Caswell, "'The Archive' Is Not an Archives."
27. Desai, "An Archival Perspective on Pretraining Data."
28. Hutchinson et al., "Towards Accountability."
29. Duranti, "The Long-Term Preservation."
30. Fernandes, "Data Preparation." See also Sancricca, "Enhancing Data Preparation."
31. See note 28 above.
32. See note 13 above.
33. See note 12 above.
34. See note 13 above.
35. GO FAIR, "FAIR Principles."
36. Li, "CultureLLM"; and Vanallemeersch et al., "AI4Culture."
37. Hiniduma et al., "Data Readiness for AI"; and "AI Data Readiness Inspector."
38. Wiedeman, "The Historical Hazards of Finding Aids."
39. Carter, "Of Things Said and Unsaid"; and Schwartz and Cook, "Archives, Records, and Power."
40. Nelson, "Archiving Hate."
41. Vanallemeersch et al., "AI4Culture."
42. See note 17 above.

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